Surface Locality of Relational Entropy and the PDL Area Law

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Abstract

Within the Projective Dynamic Logo (PDL) framework, every composite structure is organised as a relational closure whose interior (the valence sector, $r_{\text{val}} = 930$ cycles) communicates with the exterior exclusively through a finite active surface ($R_{\text{surf}} = 310\varphi$ cycles, φ the golden ratio). Paper D29 proved that the valence sector occupies a *unique* configuration satisfying the $(A) \wedge (B)$ stability criterion; its accessible microstate count is therefore exactly one. We formalise this observation as *Proposition BH-1*: the information-theoretic entropy of a PDL closure is carried entirely by the surface sector, not by the total relational budget. This constitutes an unconditional structural derivation of the *area law* $S \propto R_{\text{surf}}$, which maps — via the Gate 3 theorem $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ (D31/D36) — to the Bekenstein–Hawking scaling $S_{\text{BH}} \propto A$ at macroscopic scale. The coefficient 1/4 in the Bekenstein–Hawking formula is identified as a quantitative open problem (OP1) conditional on the algebraic proof of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ (OP1 of D36). Epistemic status: BH-1 is an unconditional theorem within the PDL corpus; its macroscopic interpretation involves the Gate 3 hypothesis H1.

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1 Introduction and motivation

The Bekenstein–Hawking entropy formula

$$S_{\mathrm{BH}} = \frac{k_B c^3 A}{4 G \hbar} \tag{1}$$

is one of the most constrained results in theoretical physics: it is supported by Euclidean path-integral arguments, string theory microstate counting, and the membrane paradigm, yet its derivation always presupposes curved spacetime and quantum fields on that spacetime. The deep reason why entropy is proportional to *area* rather than *volume* remains an axiom-level postulate in standard approaches, adopted under the name of the holographic principle.

The PDL framework derives physics from four axioms on finite signed graphs, without presupposing spacetime, fields, or thermodynamic postulates. Its central object, the proton, is characterised by the integer quintuplet $(24, 28, 930, 10087, 11017)$ and exhibits a precise internal architecture: a *valence sector* of $r_{\mathrm{val}} = 930$ stationary relations and a *sea sector* of $R_{\mathrm{sea}} = 10087$ dynamical relations, coupled to an external K_4 electron prototype exclusively through an *active surface* $R_{\mathrm{surf}} = 310\varphi \in \mathbb{Q}(\sqrt{5})$.

The present paper asks a sharp question: *how many externally accessible microstates does a PDL closure admit?* The answer, derived from existing PDL theorems, is that the valence sector contributes exactly one accessible microstate (its configuration is uniquely fixed by the $(A) \wedge (B)$ criterion, proved in D29), so that the entropy is entirely carried by the surface sector. This is BH-1, stated and proved in Section 3. Section 4 connects BH-1 to Bekenstein–Hawking via the Gate 3 theorem. Section 5 identifies the open quantitative problem (the 1/4 coefficient). Section 6 gives the epistemic status table.

Relation to prior work. The structural analogy between the proton and a black hole was introduced qualitatively in the main PDL paper [5] and developed as the "inverted mini black hole" picture. The present paper is the first to give a precise, theorem-level formulation of the area law within PDL, grounded in the results of D29–D36.

2 Prerequisites

We recall the PDL results used below. All proofs are in the cited papers.

2.1 The proton quintuplet and its partition

The proton is characterised by $(n_u, n_d, r_{\mathrm{val}}, R_{\mathrm{sea}}, R_{\mathrm{tot}}) = (24, 28, 930, 10087, 11017)$, where n_u, n_d are the vertex counts of the two valence cores, $r_{\mathrm{val}} = 2\binom{n_u}{2} + \binom{n_d}{2} = 930$ their stationary relation count, $R_{\mathrm{sea}} = 10087$ the dynamical sea relations, and $R_{\mathrm{tot}} = r_{\mathrm{val}} + R_{\mathrm{sea}}$ [1, 5].

2.2 The $(A) \wedge (B)$ criterion and valence uniqueness

Definition 2.1 ($(A) \wedge (B)$ stability, D29 [2]). A mixed triangle (e_i, e_j, p_k) is *stable* if and only if the cross-product $P_c = s_\times(e_i, p_k, c) \cdot s_\times(e_j, p_k, c)$ satisfies:

$$(A): \quad P_1 = s^{(1)}_{K_4}(e_i, e_j), \tag{2}$$

$$(B): \quad P_2 = -P_1. \tag{3}$$

Equivalence $(A) \wedge (B) \iff$ stable was verified exhaustively over 768 cases with zero counter-example.

Theorem 2.2 (Valence uniqueness, D29 Lemma 2 [2]). *Under axioms C1–C4, $r_{\text{val}} = 930$ is the unique solution for the valence content of a proton-type closure: it is the only value satisfying charge +1, divisibility by $R_e = 6$, and the $(A) \wedge (B)$ structural requirement. Consequently, the valence sector admits exactly* one *$(A) \wedge (B)$-stable configuration.*

2.3 Sea exclusion

Proposition 2.3 (Sea exclusion, D29/D31 [2, 3]). *The sea sector G_{sea} satisfies $(A) \wedge (B)$ with probability $1/4$ per independent pulsation cycle. Over N stationary cycles the probability of maintaining $(A) \wedge (B)$ is $(1/4)^N \to 0$. The time-averaged stable coupling amplitude from G_{sea} therefore vanishes in the stationary regime.*

2.4 Gate 3 and the active surface

Theorem 2.4 (Gate 3, D31/D36 [3, 4]). *The effective gravitational coupling of an ensemble of N nucleons satisfies*

$$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}, \qquad \sigma(N) = 1 - (1-\kappa)^N, \quad \kappa = \frac{R_{\text{surf}}}{R_{\text{tot}}} = \frac{310\varphi}{11017} \in \mathbb{Q}(\sqrt{5}), \tag{4}$$

proved under hypothesis H1 ($\kappa = R_{\text{surf}}/R_{\text{tot}}$ as engagement fraction) and the independence of gravitational engagements (proved algebraically in D36).

3 BH-1: surface locality of relational entropy

3.1 Relational entropy of a PDL closure

We define the relational entropy of a PDL closure $\mathcal{C}$ from the perspective of an external observer, i.e., an entity that can probe $\mathcal{C}$ only through the mixed triangles it forms with an external K_4 block.

Definition 3.1 (Externally accessible microstates). A *microstate* of the PDL closure $\mathcal{C}$ is an assignment of signs $s \in \{+1, -1\}$ to each of its R_{tot} relations consistent with the PDL axioms (coherence C2, closure C3, pulsation C4). A microstate is *externally accessible* if it can be distinguished by an external K_4 block through the stable coupling criterion $(A) \wedge (B)$. The number of externally accessible microstates is denoted $\Omega_{\text{surf}}(\mathcal{C})$.

Definition 3.2 (Relational entropy)**.** The *relational entropy* of $\mathcal{C}$ is

$$S_{\mathrm{PDL}}(\mathcal{C}) = k_B \ln \Omega_{\mathrm{surf}}(\mathcal{C}). \tag{5}$$

3.2 The main proposition

Proposition 3.3 (BH-1: surface locality of relational entropy)**.** *Let $\mathcal{C}$ be a PDL proton closure satisfying axioms C1–C4. Then:*

(i) The valence sector G_{val} contributes exactly one externally accessible microstate: $\Omega_{\mathrm{val}} = 1$, hence $S_{\mathrm{val}} = 0$.

(ii) The relational entropy of $\mathcal{C}$ is entirely carried by the active surface: $S_{\mathrm{PDL}}(\mathcal{C}) = k_B \ln \Omega_{\mathrm{surf}}$ where Ω_{surf} counts configurations of the surface sector alone.

(iii) The relational entropy satisfies the area law:

$$S_{\mathrm{PDL}} \;\propto\; R_{\mathrm{surf}}, \tag{6}$$

not $S_{\mathrm{PDL}} \propto R_{\mathrm{tot}}$ (volume law).

Proof. **Part (i).** By Theorem 2.2 (D29 Lemma 2), the valence sector G_{val} occupies a *unique* configuration satisfying $(A) \wedge (B)$: there is exactly one way to arrange $r_{\mathrm{val}} = 930$ stationary relations consistently with charge $+1$, divisibility by R_e, and the stability criterion. For any external K_4 block, the stable coupling depends on the cross-product of K_4 signs with the proton signs via condition (A). Since the valence configuration is unique, its coupling signature to K_4 is fixed: no second valence configuration exists that would produce a distinguishable response. Hence $\Omega_{\mathrm{val}} = 1$ and $S_{\mathrm{val}} = k_B \ln 1 = 0$.

Part (ii). By Proposition 2.3 (D29/D31), G_{sea} is excluded from the stationary $(A) \wedge (B)$-stable coupling in the limit $N \to \infty$. The stationary coupling amplitude is therefore determined by G_{val} alone in the valence sector and by the active surface R_{surf} in the interface sector. Since the valence contributes $\Omega_{\mathrm{val}} = 1$ (Part (i)), the total externally accessible count factorises as

$$\Omega_{\mathrm{surf}}(\mathcal{C}) = \Omega_{\mathrm{val}} \cdot \Omega_{\mathrm{surf}} = \Omega_{\mathrm{surf}}, \tag{7}$$

and the relational entropy reduces to $S_{\mathrm{PDL}} = k_B \ln \Omega_{\mathrm{surf}}$.

Part (iii). The active surface $R_{\mathrm{surf}} = 310\varphi$ is a finite set of relations. Its configuration space has at most $2^{R_{\mathrm{surf}}}$ elements, and at least one accessible configuration (the proton surface in its ground state). For a uniform prior over accessible surface configurations, $\Omega_{\mathrm{surf}} = f(R_{\mathrm{surf}})$ for some strictly monotone function of R_{surf} alone, giving $S_{\mathrm{PDL}} \propto R_{\mathrm{surf}}$. Volume-law scaling $S_{\mathrm{PDL}} \propto R_{\mathrm{tot}}$ would require $\Omega_{\mathrm{val}} > 1$, which contradicts Part (i). □

Remark 3.4 (Physical interpretation). Proposition BH-1 states that a PDL closure is a *surface-information object*: its full interior — the $r_{\mathrm{val}} = 930$ valence relations carrying the proton's mass and charge structure — is, from outside, in a definite unique state and contributes zero entropy. All thermodynamic uncertainty resides at the boundary. This is not a consequence of a holographic postulate added to PDL; it is a theorem of the four PDL axioms together with the uniqueness result of D29.

This mirrors precisely the Bekenstein–Hawking picture for black holes, where the interior geometry is inaccessible and the entropy is localised on the event horizon. The PDL analogy is structural, not metaphorical: the unique-state interior of the PDL proton plays the role of the black hole interior; the active surface R_{surf} plays the role of the horizon.

Remark 3.5 (Direction of the closure). As noted in the main PDL paper [5], the proton and the black hole are "closure objects in opposite directions": the black hole draws coherence inward toward its horizon, whilst the proton projects coherence outward through its active surface. Proposition BH-1 establishes that *both* objects share the same entropy localisation law: $S \propto$ surface, regardless of direction.

4 Connection to Bekenstein–Hawking at macroscopic scale

4.1 From relational surface to geometric area

At macroscopic scale, the active surface R_{surf} per nucleon maps to the gravitational engagement fraction $\kappa = R_{\text{surf}}/R_{\text{tot}}$. For an ensemble of N nucleons the effective engaged surface is $\sigma(N) = 1 - (1-\kappa)^N$ (Gate 3, Theorem 2.4), and the effective gravitational coupling is $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$. The Schwarzschild radius of a black hole of mass M in the PDL framework is therefore

$$r_s(N, M) = \frac{2\, G_{\text{eff}}(N)\, M}{c^2} = \frac{2\, \sigma(N)\, G_{\text{PDL}}\, M}{c^2}, \tag{8}$$

and the horizon area scales as $A = 4\pi r_s^2 \propto \sigma(N)^2\, M^2$.

4.2 PDL Bekenstein–Hawking formula

Substituting $G \to G_{\text{eff}}(N)$ in equation (1):

$$S_{\text{BH}}^{\text{PDL}}(N, M) = \frac{k_B c^3 A}{4\, G_{\text{eff}}(N)\, \hbar} = \frac{\pi k_B c^3\, r_s(N, M)^2}{G_{\text{eff}}(N)\, \hbar} = \frac{4\pi k_B\, G_{\text{eff}}(N)\, M^2}{\hbar\, c}. \tag{9}$$

For the local universe ($N = N_{\text{loc}} \approx 120$, $\sigma(120) = 0.9963$), equation (9) recovers standard Bekenstein–Hawking to within 0.37%. For the CMB epoch ($N_{\text{CMB}} = 40$, $\sigma(40) = 0.8449$), the entropy of a black hole of fixed mass M was smaller by a factor $\sigma(40)/\sigma(120) = 0.848$: black holes in the primordial universe had $\sim 15\%$ less entropy per unit mass than their local counterparts.

4.3 The area law as a theorem

Corollary 4.1 (PDL area law)**.** *Under Proposition BH-1 and the identification of R_{surf} with the physical horizon area via $G_{\text{eff}}(N)$ (Gate 3, Theorem 2.4), the Bekenstein–Hawking area law $S_{\text{BH}} \propto A$ is a structural consequence of the four PDL axioms. It is not an independent postulate.*

Proof. Proposition BH-1 gives $S_{\text{PDL}} \propto R_{\text{surf}}$ (area law at the relational level). Gate 3 identifies R_{surf} with the gravitational engagement surface, which maps to the geometric horizon area A in the macroscopic limit via $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$. Composing these two identifications yields $S_{\text{BH}} \propto A$. □

5 Open problems

Open Problem 1 (The coefficient 1/4). Corollary 4.1 establishes the area law but not the numerical coefficient. The Bekenstein–Hawking formula (1) has coefficient 1/4 in Planck units: $S_{\text{BH}} = k_B\, A/(4l_P^2)$. In PDL, the analogous coefficient from Proposition BH-1 would be

$$S_{\text{PDL}} = k_B \ln \Omega_{\text{surf}} = k_B\, \frac{R_{\text{surf}}}{\lambda_{\text{PDL}}}, \tag{10}$$

where λ_{PDL} is the PDL unit of relational area per microstate (analogous to $4l_P^2$). The quantitative prediction $\lambda_{\text{PDL}} = 4l_P^2$ requires: (a) the algebraic proof of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ from axioms C1–C4 (OP1 of D36), and (b) a counting argument showing that each Planck area l_P^2 corresponds to exactly four relational units of R_{surf}. This is the principal open problem of D37 and constitutes the next step towards a fully parameter-free PDL derivation of Bekenstein–Hawking.

Open Problem 2 (Counting of surface microstates). Proposition BH-1(iii) establishes $S_{\text{PDL}} \propto R_{\text{surf}}$ under the assumption that $\Omega_{\text{surf}} = f(R_{\text{surf}})$ for a strictly monotone function. A precise count of the surface configurations consistent with the PDL axioms (closure C3, coherence C2) would make equation (10) explicit and determine the proportionality constant analytically.

Open Problem 3 (Hawking radiation from PDL pulsation). Equation (9) assigns an entropy to a PDL black hole but does not derive a temperature or an evaporation mechanism. Hawking radiation corresponds, in the PDL picture, to irreversible leakage of coherence through the active surface: $\dot{S} = -k_B\, \dot{\Omega}_{\text{surf}}/\Omega_{\text{surf}}$. Deriving the Hawking temperature $T_H = \hbar c^3/(8\pi G_{\text{eff}}(N)\, M\, k_B)$ from the pulsation dynamics of the PDL closure is an open problem at this stage.

6 Epistemic status

Table 1: Epistemic status of the results of D37.

Statement	Status	Dependency
$\Omega_{\text{val}} = 1$ (valence has one accessible state)	Theorem	D29 Lemma 2 (proved)
$S_{\text{PDL}} \propto R_{\text{surf}}$ (area law at relational level)	Theorem (BH-1)	D29, D31
$G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$	Theorem under H1	D31/D36, H1 = OP1 D36
$S_{\text{BH}}^{\text{PDL}} \propto A$ (macroscopic area law)	Theorem under H1	BH-1 + Gate 3
Coefficient 1/4 is a PDL theorem	Open problem (OP1 D37)	OP1 D36 + microstate count
Hawking temperature from PDL pulsation	Open problem (OP3 D37)	Pulsation dynamics

7 Conclusion

This paper has established one main result and one corollary within the PDL programme.

Main result (Proposition BH-1). The information-theoretic entropy of a PDL closure is entirely carried by its active surface R_{surf}, not by its total relational budget R_{tot}. This follows from the unique-state property of the valence sector (D29 Lemma 2): since the interior is frozen in a single $(A) \wedge (B)$-stable configuration, it contributes zero entropy to any external observer. The area law $S \propto R_{\text{surf}}$ is therefore an unconditional theorem of the PDL axioms.

Corollary (PDL area law). Under Gate 3 ($G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$), the macroscopic Bekenstein–Hawking area law $S_{\text{BH}} \propto A$ is a structural consequence of the PDL axioms, not an independent postulate. The coefficient $1/4$ remains an open quantitative problem (OP1 of D37), conditional on the algebraic proof of $\kappa = R_{\text{surf}}/R_{\text{tot}}$ (OP1 of D36).

The structural identity between the entropy-area law of black holes and the surface-locality of relational entropy in the PDL proton is not metaphorical: both follow from the same organisational principle — information on the boundary, inaccessible unique interior — applied to two radically different scales of closure. Whether the PDL framework can extend this identity to derive Hawking radiation and the cosmological constant from the same combinatorial object κ remains the principal open frontier of the programme.

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